

Academic Appointments

2025 – **Massachusetts Institute of Technology**
present Postdoctoral Associate in Aeronautics and Astronautics
Advisor: Prof. Julie Shah

Education

2018 – 2025 **Brown University**
Ph.D. in Computer Science
Advisor: Prof. Stefanie Tellex

2021 **Brown University**
M.S. in Computer Science
Advisor: Prof. Stefanie Tellex

2017 **University of California, Berkeley**
B.S. in Electrical Engineering and Computer Sciences
Advisor: Prof. Ken Goldberg and Prof. David Wagner

Fellowships, Awards and Honors

Brown **National Science Foundation Graduate Research Fellowship Program**, 2018
Jack Kent Cooke Foundation Graduate Scholarship, 2018

UC Berkeley **Tau Beta Pi, National Engineering Honor Society**, 2014
Eta Kappa Nu, National EECS Honor Society, 2014
Jack Kent Cooke Foundation Undergraduate Transfer Scholarship, 2014

Publications

CoRL 2026 **Language as a Sensor: Calibrated Spatial Belief Estimation in 3D Scenes from Natural Language**
A. Naveen, **J.X. Liu**, L. Carlone, A. Bobu
Conference on Robot Learning (CoRL) 2026. [Paper]

HRI 2026 **LEGS-POMDP: Language and Gesture-Guided Object Search in Partially Observable Environments**
I. He, S. Tellex, **J.X. Liu**
ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026. [Paper, Website]

RA-L 2025 **Open-vocabulary Pick and Place via Patch-level Semantic Maps**
M. Jia, H. Huang, Z. Zhang, C. Wang, L. Zhao, D. Wang, **J.X. Liu**, R. Walters, R. Platt, S. Tellex
IEEE Robotics and Automation Letters (RA-L) 2025. **Best Paper Award Honorable Mention**. [Paper, Website]

IROS 2024 **Lang2LTL-2: Grounding Spatiotemporal Navigation Commands Using Large Language and Vision-Language Models**
J.X. Liu, A. Shah, G. Konidaris, S. Tellex, and D. Paulius
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024. [Paper, Website]

- IJCAI 2024 **A Survey of Robotic Language Grounding: Tradeoffs between Symbols and Embeddings**
V. Cohen*, **J.X. Liu***, R. Mooney*, S. Tellex*, D. Watkins*
International Joint Conference on Artificial Intelligence (IJCAI) 2024 Survey Track. [Paper]
- ICRA 2024 **LTL-Transfer: Skill Transfer for Temporally-Extended Task Specifications**
J.X. Liu*, A. Shah*, E. Rosen, M. Jia, G.D. Konidaris, S. Tellex
IEEE International Conference on Robotics and Automation (ICRA) 2024. [Paper, Website]
- CoRL 2023 **Lang2LTL: Grounding Complex Natural Language Commands for Temporal Tasks in Unseen Environments**
J.X. Liu*, Z. Yang*, I. Idrees, S. Liang, B. Schornstein, S. Tellex, A. Shah
Conference on Robot Learning (CoRL) 2023. [Paper, Website, Brown News]
AAAI Fall Symposium on Unifying Representations for Robot Application Development (UR-RAD) 2023. **Best Paper Finalist.**
- INTERSPEECH 2023 **Style-transfer based Speech and Audio-visual Scene Understanding for Robot Action Sequence Acquisition from Videos**
C. Hori, P. Peng, D. Harwath, **J.X. Liu**, K. Ota, S. Jain, R. Corcodel, D. Jha, D. Romeres, J. Le Roux
INTERSPEECH 2023. [Paper]
- ICRA 2022 **Generalizing to New Domains by Mapping Natural Language to Lifted LTL**
E. Hsiung, H. Mehta, J. Chu, **J.X. Liu**, R. Patel, S. Tellex, G. D. Konidaris
IEEE International Conference on Robotics and Automation (ICRA) 2022. [Paper]
- ACNS 2020 **Minority Reports Defense: Defending Against Adversarial Patches**
M. McCoyd, W. Park, S. Chen, N. Shah, R. Roggenkemper, M. Hwang, **J.X. Liu**, D. Wagner
Applied Cryptography and Network Security Workshops (ACNS) 2020. [Paper]
- ICRA 2018 **Dex-Net 3.0: Computing Robust Vacuum Suction Grasp Targets in Point Clouds Using a New Analytic Model and Deep Learning**
J. Mahler, M. Matl, **J.X. Liu**, A. Li, D. Gealy, K. Goldberg
IEEE International Conference on Robotics and Automation (ICRA) 2018. [Paper, Website]
- ICISS 2018 **Detecting Phone Theft Using Machine Learning**
J.X. Liu, D. Wagner, S. Egelman
International Conference on Information Science and System (ICISS) 2018. [Paper]
- RSS 2017 **Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics**
J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, **J.X. Liu**, J. A. Ojea, K. Goldberg
Robotics: Science and Systems (RSS) 2017. [Paper, Website]

Dissertation

- August 2025 **Robotic Language Grounding**
Jason Xinyu Liu
Brown Computer Science 2025. [Thesis, Video]

Invited Talks

- March 2026 The University of Edinburgh, RL Seminar
March 2026 Oxford, Robotics Seminar
Oct 2025 UIUC, CS Colloquium
Oct 2025 Tufts, HRI Seminar
Jan 2025 USC, Robotics Seminar
Dec 2024 Brown, Brown Database Group

Oct 2024 Amazon Robotics, Amazon Robotics Research Fall Symposium and PhD Communication Competition

Oct 2024 UT Austin, Forum for Artificial Intelligence

Oct 2024 CU Boulder, CS Colloquium

Sep 2024 MIT, Interactive Robotics Group

Sep 2024 UT Austin, Center of Autonomy

Aug 2024 Arizona State, Logos Robotics Lab

Jul 2024 Oxford, CS Department Seminar

Oct 2022 Northeast Robotics Colloquium

Guest Lectures

Spring 2026 Sapienza University of Rome, Hybrid Neurosymbolic Techniques for Non-Markovian Reinforcement Learning

Spring 2025 Brown, CSCI 1470: Deep Learning

Fall 2024 Brown, CSCI 1951X: Formal Proof and Verification

Workshop Organization

IROS 2026 Leveraging Human-Robot Dialogue to Improve Robot Learning and Human-Robot Interaction. [Website]

IROS 2026 Learning and Formal Methods for Robotics. [Website]

CoRL 2025 Evaluation and Deployment Across the Robot Learning Lifecycle. [Website]

RSS 2025 Robot Planning in the Era of Foundation Models. [Website]

RSS 2024 Workshop on Task Specification for General-Purpose Intelligent Robots. [Website]

Industry Experience

Summer 2022 **Research Intern**, *Mitsubishi Electric Research Laboratories*, Cambridge, MA

Summer 2015 **Software Engineer**, *NetSuite*, San Mateo, CA

Mentorship

2026 - Now Aryan Naveen (MIT Ph.D.)

2026 Summer Jonas Xiangzhou Sun (MIT Undergraduate)

2026 Summer Jefferson Lin (MIT Undergraduate)

2026 Summer Amanda Cheng (Wellesley Undergraduate)

2024 - 2025 Ivy He (Brown Ph.D.)

2023 - 2025 Yichen Wei (Brown Master's. Now Ph.D. at UPenn)

2023 - 2024 Anneke Wernerfelt (Haverford Undergraduate -> Master's at UPenn)

Spring 2021 Shihab Khalfalla (UConn Undergraduate. Now at Microsoft)

2021 - 2022 Eric Hsiung (Brown Master's. Now Ph.D. at UT Austin)

2020 - 2021 Tyler Edwards (UPenn Undergraduate -> Master's at UPenn)

2020 - 2021 Suchen Zheng (Brown Master's. Now Ph.D. at Yale)

Service

Brown **First-year Ph.D. student mentor**, 2021-2023

Google exploreCSR (explorecsr.cs.brown.edu), Spring 2021

Review **THRI** 2026, **RSS** 2026, **JMLR** 2025, **CoRL-W** 2024, **RSS-W** 2024, **IROS** 2024, **ICRA** 2023,
ICRA 2022